

Determine whether the series converges conditionally, converges absolutely, or diverges.

1.

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln(n)}$$

☐ Diverges

☐ Converges conditionally

☐ Converges absolutely

Reason:

2.

$$\sum_{n=0}^{\infty} \cos(\pi n) \frac{n+1}{n+3}$$

☐ Diverges

☐ Converges conditionally

☐ Converges absolutely

Reason:

3.

$$\sum_{n=1}^{\infty} \frac{n}{5^n}$$

☐ Diverges

☐ Converges conditionally

☐ Converges absolutely

Reason:

4.

$$\sum_{n=1}^{\infty} \frac{2^n \ln(n)}{n!}$$

☐ Diverges

☐ Converges conditionally

☐ Converges absolutely

Reason:

5.

$$\sum_{n=1}^{\infty} \frac{n^n}{n!}$$

☐ Diverges

☐ Converges conditionally

☐ Converges absolutely

Reason: